In [3]:
```
import pandas as pd
import numpy as np

df = pd.DataFrame({
    "Mission_ID": [
        "M001", "M002", "M003", "M004", "M005",
        "M006", "M007", "M008", "M009", "M010",
        "M011", "M012", "M013", "M014", "M015",
        "M016", "M017", "M018", "M019", "M020"
    ],

    "Mission_Name": [
        "Mars Observer", "lunar explorer", "Voyager Probe",
        "Mars Observer", "Europa Clipper", "Chandrayaan-3",
        "SATURN EXPLORER", "Jupiter Orbiter", "Lunar Explorer",
        "Mars Rover", "Neptune Mission", "Solar Probe",
        "Comet Tracker", "Asteroid Survey", "Mars Rover",
        "Deep Space Probe", "Venus Orbiter", "Mercury Explorer",
        "ISS Research", "Pluto Explorer"
    ],

    "Observation_Date": [
        "2026-01-05", "2026-01-08", "2026-01-12",
        "2026-01-05", "2026-01-15", "2026-01-18",
        "2026-01-20", "2026-01-22", "2026-01-25",
        "2026-01-28", "2026-02-01", "2026-02-04",
        "2026-02-07", "2026-02-10", "2026-02-13",
        "2026-02-16", "2026-02-19", "2026-02-22",
        "2026-02-25", "invalid-date"
    ],

    "Region": [
        "Mars", "Moon", "Outer Solar System", "Mars",
        "Jupiter", "Moon", "Saturn", "Jupiter", "Moon",
        "Mars", "Outer Solar System", "Sun", "Asteroid Belt",
        "Asteroid Belt", "Mars", "Outer Solar System",
        "Venus", "Mercury", "Earth Orbit", "Outer Solar System"
    ],

    "Distance_Mkm": [
        225.4, 0.384, 24000, 225.4, 628,
        0.384, 1427, 778.5, 0.384, 225.4,
        4500, 0.7, 35.6, 12.8, 225.4,
        999999, 41.4, 77.3, 0.0004, "unknown"
    ],

    "Signal_Strength": [
        85, 92, 45, 85, 76, 95, 68, np.nan, 91, 82,
        39, 105, 72, 67, 88, 25, -10, 81, 89, 55
    ],

    "Data_Collected_GB": [
        12.5, 8.2, 2.8, 12.5, 15.4,
        10.1, 7.6, 18.3, np.nan, 11.7,
```

```
        3.2, 22.5, 6.4, 5.9, 1500,
        1.5, 9.8, 0, 14.2, 4.7
    ],

    "Status": [
        "Active", "Active", "Active", "Active", "In Transit",
        "Completed", "Active", "Active", "Active", "active",
        "In Transit", "Active", "Completed", "Unknown", "Active",
        "In Transit", "Completed", "Active", " Active", "Completed"
    ]
})

df
```

Out[3]:

	Mission_ID	Mission_Name	Observation_Date	Region	Distance_Mkm	Signal_Strength
0	M001	Mars Observer	2026-01-05	Mars	225.4	85.0
1	M002	lunar explorer	2026-01-08	Moon	0.384	92.0
2	M003	Voyager Probe	2026-01-12	Outer Solar System	24000	45.0
3	M004	Mars Observer	2026-01-05	Mars	225.4	85.0
4	M005	Europa Clipper	2026-01-15	Jupiter	628	76.0
5	M006	Chandrayaan-3	2026-01-18	Moon	0.384	95.0
6	M007	SATURN EXPLORER	2026-01-20	Saturn	1427	68.0
7	M008	Jupiter Orbiter	2026-01-22	Jupiter	778.5	NaN
8	M009	Lunar Explorer	2026-01-25	Moon	0.384	91.0
9	M010	Mars Rover	2026-01-28	Mars	225.4	82.0
10	M011	Neptune Mission	2026-02-01	Outer Solar System	4500	39.0
11	M012	Solar Probe	2026-02-04	Sun	0.7	105.0
12	M013	Comet Tracker	2026-02-07	Asteroid Belt	35.6	72.0
13	M014	Asteroid Survey	2026-02-10	Asteroid Belt	12.8	67.0
14	M015	Mars Rover	2026-02-13	Mars	225.4	88.0
15	M016	Deep Space Probe	2026-02-16	Outer Solar System	999999	25.0
16	M017	Venus Orbiter	2026-02-19	Venus	41.4	-10.0
17	M018	Mercury Explorer	2026-02-22	Mercury	77.3	81.0
18	M019	ISS Research	2026-02-25	Earth Orbit	0.0004	89.0
19	M020	Pluto Explorer	invalid-date	Outer Solar System	unknown	55.0

In [5]:
```
df.isnull()
```

Out[5]:

	Mission_ID	Mission_Name	Observation_Date	Region	Distance_Mkm	Signal_Strength
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
5	False	False	False	False	False	False
6	False	False	False	False	False	False
7	False	False	False	False	False	True
8	False	False	False	False	False	False
9	False	False	False	False	False	False
10	False	False	False	False	False	False
11	False	False	False	False	False	False
12	False	False	False	False	False	False
13	False	False	False	False	False	False
14	False	False	False	False	False	False
15	False	False	False	False	False	False
16	False	False	False	False	False	False
17	False	False	False	False	False	False
18	False	False	False	False	False	False
19	False	False	False	False	False	False

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
Mission_ID          0
Mission_Name        0
Observation_Date     0
Region              0
Distance_Mkm        0
Signal_Strength     1
Data_Collected_GB   1
Status              0
dtype: int64
```

In [9]:
```
df.isnull().sum()[df.isnull().sum() > 0]
```

Out[9]:
```
Signal_Strength     1
Data_Collected_GB   1
dtype: int64
```

In [11]:
```
df.duplicated()
```

Out[11]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18    False
19    False
dtype: bool
```

In [13]:
```
df.duplicated().sum()
```

Out[13]:
```
np.int64(0)
```

In [15]:
```
df = pd.concat([df, df.iloc[[0]]], ignore_index=True)
```

In [17]:
```
df.duplicated()
```

Out[17]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
16    False
17    False
18    False
19    False
20     True
21     True
dtype: bool
```

In [21]:
```
df.duplicated().sum()
```

Out[21]:
```
np.int64(2)
```

In [23]:
```
df[df.duplicated(keep=False)]
```

Out[23]:

	Mission_ID	Mission_Name	Observation_Date	Region	Distance_Mkm	Signal_Strength
0	M001	Mars Observer	2026-01-05	Mars	225.4	85.0
20	M001	Mars Observer	2026-01-05	Mars	225.4	85.0
21	M001	Mars Observer	2026-01-05	Mars	225.4	85.0

In [30]:
```
df = df.drop_duplicates()
df.duplicated().sum()
```

Out[30]:
```
np.int64(0)
```

In [33]:
```
## Highest valid distance

df["Distance_Mkm"] = pd.to_numeric(
    df["Distance_Mkm"], errors="coerce"
)

df["Distance_Mkm"]
```

Out[33]:
```
0        225.4000
1          0.3840
2      24000.0000
3        225.4000
4        628.0000
5          0.3840
6       1427.0000
7        778.5000
8          0.3840
9        225.4000
10      4500.0000
11         0.7000
12        35.6000
13        12.8000
14       225.4000
15    999999.0000
16        41.4000
17        77.3000
18         0.0004
19            NaN
Name: Distance_Mkm, dtype: float64
```

In [35]:
```
highest_distance = df["Distance_Mkm"].max()
print(highest_distance)
```

```
999999.0
```

In [37]:
```
## group by region
```

```
df["Data_Collected_GB"].sum()
```

Out[37]: np.float64(1667.3)

In [39]:
```
df.groupby("Region")["Data_Collected_GB"].sum()
```

Out[39]:
```
Region
Asteroid Belt          12.3
Earth Orbit            14.2
Jupiter                33.7
Mars                 1536.7
Mercury                 0.0
Moon                   18.3
Outer Solar System     12.2
Saturn                  7.6
Sun                    22.5
Venus                   9.8
Name: Data_Collected_GB, dtype: float64
```

In [41]:
```
df.groupby("Region")["Data_Collected_GB"].sum().sort_values(
    ascending=False
)
```

Out[41]:
```
Region
Mars                 1536.7
Jupiter                33.7
Sun                    22.5
Moon                   18.3
Earth Orbit            14.2
Asteroid Belt          12.3
Outer Solar System     12.2
Venus                   9.8
Saturn                  7.6
Mercury                 0.0
Name: Data_Collected_GB, dtype: float64
```

In [42]:
```
df.groupby("Region")["Data_Collected_GB"].max()
```

Out[42]:
```
Region
Asteroid Belt          6.4
Earth Orbit           14.2
Jupiter               18.3
Mars                1500.0
Mercury                0.0
Moon                  10.1
Outer Solar System     4.7
Saturn                 7.6
Sun                   22.5
Venus                  9.8
Name: Data_Collected_GB, dtype: float64
```